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IMAGE FORMING SYSTEM FOR INSPECTING QUALITY OF IMAGE FORMED ON SHEET AND GENERATING CORRECTION DATA TO CORRECT SECOND IMAGE USING THE CORRECTION DATA

Abstract

An image forming system generates second image data representing a result of a reading unit reading an image on a sheet, corrects the second image data using correction data, the correction data being data obtained on the basis of image data obtained by the reading unit reading a predetermined image formed on the sheet by an image forming unit and image data corresponding to the predetermined image, and inspects, on the basis of the corrected second image data and a first image data, an image formed on the sheet on the basis of the first image data.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an image forming system for inspecting the quality of an image formed on a sheet.

Description of the Related Art

[0002] A known system uses an inspection apparatus connected to an image forming apparatus for forming an image on a sheet to read an image on a sheet conveyed from the image forming apparatus and inspect the image on the sheet on the basis of the read image. In US-2021-0073966, a RIP image and an image obtained by reading an image formed on a sheet on the basis of the RIP image are compared and the quality (print quality) of the image formed on the sheet is inspected. RIP is an acronym for raster image processing.

[0003] A system for reading an image on a conveyed sheet includes a first conveyance unit that conveys a sheet when an image on the sheet is being read and a second conveyance unit that passes the sheet to the first conveyance unit. In a case in which a motor that drives the first conveyance unit is a motor different from a motor that drives the second conveyance unit, the speed at which the sheet is conveyed during image reading may vary. In other words, when the conveyance speed of the sheet varies while the image is being read by an inspection apparatus, the image (image for reading) representing the reading result may be elongated or shortened in the direction corresponding to the conveyance direction of the sheet. When print quality is inspected on the basis of a comparison between such an image for reading and a RIP image, the inspection accuracy may be decreased.

SUMMARY OF THE INVENTION

[0004] The present disclosure provides an image forming system comprising: an image forming unit configured to form an image corresponding to first image data input into the image forming system on a sheet; a first conveyance unit configured to convey the sheet conveyed from the image forming unit; a first motor configured to drive the first conveyance unit; a second conveyance unit configured to convey the sheet conveyed from the first conveyance unit; a second motor configured to drive the second conveyance unit; a reading unit configured to read an image on the sheet being conveyed by the second conveyance unit; and one or more processors configured to: generate second image data representing a result of the reading unit reading an image on the sheet; correct the second image data using correction data, the correction data being data obtained on the basis of image data obtained by the reading unit reading a predetermined image formed on the sheet by the image forming unit and image data corresponding to the predetermined image; and inspect, on the basis of the corrected second image data and the first image data, an image formed on the sheet on the basis of the first image data.

[0005] Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a diagram for describing an image forming system.

[0007] FIG. 2 is a diagram for describing a controller.

[0008] FIG. 3 is a diagram illustrating an example of a profile.

[0009] FIG. 4 is a diagram for describing a test image.

[0010] FIG. 5 is a flowchart illustrating a method for creating a correction profile.

[0011] FIG. 6 is a diagram for describing an image inspection system.

[0012] FIG. 7 is a flowchart illustrating a method for creating a correction profile.

[0013] FIGS. 8A and 8B are diagrams for describing management information.

DESCRIPTION OF THE EMBODIMENTS

[0014] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

First Embodiment

Image Forming System

[0015] As illustrated in FIG. 1, an image forming system **100** includes an image inspection apparatus **110** and a sheet conveying apparatus (an image forming apparatus **120** and a post-processing apparatus **130**). In the conveyance direction of a sheet P, the image forming apparatus **120** is connected to the upstream side of the image inspection apparatus **110**, and the post-processing apparatus **130** is connected to the downstream side of the image inspection apparatus **110**. In other words, a housing **52** of the image forming apparatus **120** and a housing **53** of the post-processing apparatus **130** are different from a housing **51** of the image inspection apparatus **110**.

[0016] The image forming apparatus **120** forms an image on the sheet P via electrophotography. Note that the image forming apparatus **120** may form an image using another image forming method such as the inkjet method or the like.

[0017] Image forming units **20a**, **20b**, **20c**, and **20d** form yellow Y, magenta M, cyan C, and black K toner images, respective. Hereinafter, a plurality of components that are the same or similar are given the same reference numeral or sign. When distinguishing between the plurality of components, a letter of the alphabet is added to the end of the reference numeral or sign. When describing common items relating to a plurality of components, the letter of the alphabet is not added to the end of the reference numeral or sign.

[0018] A photosensitive drum **21** is an image carrier that carries an electrostatic latent image or a toner image. A charging device **22** charges the surface of the photosensitive drum **21**. An exposure device **23** forms an electrostatic latent image by irradiating the surface of the photosensitive drum **21** with a laser beam corresponding to an image signal. A developing device **24** develops the electrostatic latent image using the toner to form a toner image. A primary transfer roller **25** transfers the toner image from the photosensitive drum **21** to an intermediate transfer belt **27**. The intermediate transfer belt **27** conveys the toner image to a secondary transfer device **28**.

[0019] A feeding apparatus **1** feeds the sheet P held in the sheet housing case to the conveyance path. A conveying apparatus **2** conveys the sheet P to a registration apparatus **3**. The registration apparatus **3** corrects skew in the sheet P and conveys the sheet P to the secondary transfer device

28.

[0020] The secondary transfer device **28** transfers the toner image from the intermediate transfer belt **27** to the sheet P. A fixing device **29** fixes the toner image to the sheet P by applying heat and pressure to the toner image and the sheet P. A discharge conveying apparatus **4** conveys the sheet P with the toner image affixed to the outside of the image forming apparatus **120** and discharges the sheet P. In FIG. **1**, the sheet P is being conveyed to the image inspection apparatus **110**.

[0021] The image inspection apparatus **110** is an image reading apparatus that reads the sheet P where the image forming apparatus **120** formed an image and that was discharged from the image forming apparatus **120** and acquires an image for reading (image for inspection). When a sheet sensor **17a** detects the leading end of the sheet P, an inlet roller **15** rotates, and the sheet P is drawn in from the image forming apparatus **120** to the image inspection apparatus **110**. The inlet roller **15**, an upper backing roller **13**, a lower backing roller **14**, and an outlet roller **16** also rotate, conveying the sheet P. An image reading unit **11a** includes an image sensor that reads a first surface of the sheet P and a lighting device. An image reading unit **11b** includes an image sensor that reads a second surface of the sheet P and a lighting device. The upper backing roller **13** regulates the passage position of the sheet P by maintaining a distance within a predetermined range between the image reading unit **11a** and the sheet P. The lower backing roller **14** regulates the passage position of the sheet P by maintaining a distance within a predetermined range between the image reading unit **11b** and the sheet P. When a sheet sensor **17b** detects the trailing end of the sheet P, the inlet roller **15**, the upper backing roller **13**, the lower backing roller **14**, and the outlet roller **16** stop rotating. Note that in the present embodiment, a motor M1 that drives a conveyance roller **4a** is a motor different from a motor M2 that drives the inlet roller **15**. The conveyance roller **4a** is a conveyance unit included in the discharge conveying apparatus **4**.

[0022] In FIG. **1**, the post-processing apparatus **130** is connected to the downstream side of the image inspection apparatus **110**. A conveying apparatus **30** conveys the sheet P via a plurality of conveyance rollers **31** and switches between conveyance paths for the sheet P via path switching mechanisms **32a** and **32b** to discharge the sheet P to one of the discharge trays **34a**, **34b**, and **34c**. For example, the sheet P having passed image inspection may be discharged to the discharge tray **34b**, and the sheet P having not passed image inspection may be discharged to the discharge tray **34a**.

Control Configuration

[0023] FIG. **2** is a diagram for describing the control configuration of the image forming system **100**. Note that it is sufficient that the functions described below are implemented by at least one processor.

[0024] As illustrated in FIG. **2**, the image forming system **100** includes an external controller **200**, internal controllers **210a**, **210b**, and **210c**, and a console unit (operation panel) **260**. The external controller **200** and the internal controller **210a** are connected via a serial communication line **221a** and an image communication line **222a**. The external controller **200** and the internal controller **210b** are connected via a serial communication line **221b** and an image communication line **222b**. The external controller **200** and the internal controller **210c** are connected via a serial communication line **221c**.

[0025] The external controller **200** is a control circuit that comprehensively controls the entire image forming system **100**. A CPU **201** of the external controller **200** implements various functions in accordance with a control program stored in a read-only memory (ROM) area of a memory **202**. The memory **202** may include a random-access memory (RAM), a hard disk drive (HDD), a solid state drive (SSD), or the like. The external controller **200** may be provided inside either the image inspection apparatus **110**, the image forming apparatus **120**, or the post-processing apparatus **130**.

[0026] For example, the external controller **200** executes a predetermined processing on the image data corresponding to the image (for example, RIP image) received from the apparatus (PC, smart phone, tablet terminal, or the like) located outside of the image forming system **100** and outputs

post-predetermined-processing image data to the internal controller **210b** via the image communication line **222b**.

[0027] The internal controller **210b** is a controller that controls the image forming apparatus **120**. The internal controller **210b** may be disposed inside the image forming apparatus **120**. A memory **212a** may include a ROM, a RAM, a HDD, an SSD, or the like. A CPU **211b** controls the feeding apparatus **1**, the conveying apparatus **2**, the registration apparatus **3**, the image forming units **20**, the fixing device **29**, the discharge conveying apparatus **4**, and the like in accordance with a control program stored in the ROM area of a memory **212b** on the basis of an instruction from the external controller **200**. A CPU **211a** controls the image forming apparatus **120** so that an image corresponding to the image data received from the external controller **200** is formed on the sheet P.

[0028] The internal controller **210a** is a controller that controls the image inspection apparatus **110**. The internal controller **210a** may be disposed inside the image inspection apparatus **110**. The CPU **211a** of the internal controller **210a** implements various functions in accordance with a control program stored in the ROM area of the memory **212a**. For example, the CPU **211a** determines the position of the sheet P on the basis of the detection result from the sheet sensors **17a** and **17b**. Also, the CPU **211a** controls the motor M2 to drive the inlet roller **15**, the upper backing roller **13**, the lower backing roller **14**, and the outlet roller **16**. Furthermore, the CPU **211a** controls the image reading units **11a** and **11b** to acquire an image for inspection of the sheet P. The memory **212a** may include a RAM, a HDD, an SSD, or the like. The CPU **211a** transmits the image data of the image for inspection to the external controller **200** via the image communication line **222a**. The external controller **200** uses a correction profile described below to correct the image data of the image for inspection received from the CPU **211a**. The external controller **200** performs image inspection on the basis of the post-correction image data and the image data of the image (for example, RIP image) received from the apparatus outside of the image forming system **100** as a correct image. The CPU **211a** receives a read image command, an inspect image command, and the like from the external controller **200** via the serial communication line **221a**.

[0029] Note that the image inspection is performed as follows. For example, the inspection matter may be set to “misalignment”. The CPU **201** may make a determination of pass in a case in which the misalignment amount between the position of the correct image and the position of the image for inspection is equal to or less than a determination threshold. The CPU **201** may make a determination of fail in a case where the misalignment amount is greater than the determination threshold. Also, the inspection matter may be set to “black dot”. The CPU **201** may make a determination of pass in a case in which there are no black dots in the correct image and the size (for example, area) of a black dot in the image for inspection is equal to or less than a determination threshold. The CPU **201** may make a determination of fail in a case in which the size of the black dot is greater than the determination threshold. Note that in the present embodiment, “misalignment” and “black dot” are used as inspection matters, but these are simply examples. In the present embodiment, in a case in which the inspection matter is “misalignment”, an inspection of the relative positions between the correct image and the image for inspection is performed. However, this is merely an example. For example, the absolute position relative to an edge of the sheet P of the image for inspection may be inspected. In this case, a determination of pass is made in a case in which the distance between the absolute position of the correct image and the absolute position of the image for inspection is equal to or less than a determination threshold. A determination of fail is made in a case in which the distance is greater than the determination threshold.

[0030] The internal controller **210c** is a controller that controls the post-processing apparatus **130**. The internal controller **210c** may be disposed inside the post-processing apparatus **130**. A memory **212c** may include a ROM, a RAM, a HDD, an SSD, or the like. A CPU **211c** controls the conveyance rollers **31** and the path switching mechanisms **32a** and **32b** in accordance with a control program stored in the ROM area of the memory **212c**. The CPU **211c** receives a switch

command for the path switching mechanisms **32a** and **32b** and the like from the external controller **200** via the serial communication line **221c**.

[0031] The communication method between the external controller **200** and the internal controllers **210a**, **210b**, and **210c** used in this example is merely an example. The communication method used may be a controller area network (CAN) method. Also, the external controller **200** and the internal controllers **210a**, **210b**, and **210c** may be connected via a wired connection or by a wireless connection.

[0032] The configuration for reading an image on a conveyed sheet includes a first conveyance unit (for example, the inlet roller **15**) that conveys the sheet P when an image on the sheet P is being read and a second conveyance unit (conveyance roller **4a**) that passes the sheet P to the first conveyance unit. In a case in which the motor (for example, the motor M2) that drives the first conveyance unit is a motor different from the motor (for example, the motor M1) that drives the second conveyance unit, the speed at which the sheet is conveyed during image reading may vary. For example, in a case in which the housing **51** of the image inspection apparatus **110** and the housing **52** of the image forming apparatus **120** are different, the motor that drives the first conveyance unit and the motor that drives the second conveyance unit are different. As a result, when the conveyance speed of the sheet P varies while the image is being read by the image inspection apparatus **110**, the image (image for reading) representing the reading result may be elongated or shortened in the direction corresponding to the conveyance direction in which the sheet P is conveyed.

[0033] FIG. **3** is a diagram illustrating variation (profile **501**) in the image for reading. In FIG. **3**, the vertical axis represents the elongation amount of the image. In FIG. **3**, the horizontal axis represents the position of the image in the sub-scan direction. Note that “+” on the vertical axis corresponds to elongation of the image in the sub-scan direction, and “-” on the vertical axis corresponds to shortening of the image in the sub-scan direction.

[0034] As illustrated in FIG. **3**, the image is periodically elongated and shortened in the sub-scan direction according to the position in the sub-scan direction. This is caused by eccentricity in the conveyance roller. Also, as illustrated in FIG. **3**, spasmodic elongation and shortening occurs in the image in the sub-scan direction. This is caused by the conveyance speed of the sheet when the sheet is discharged by the image forming apparatus **120** being different from the conveyance speed of the sheet when the sheet is taken in and conveyed by the image inspection apparatus **110**. The elongation and shortening of the image is also caused by the shock that occurs when the sheet P is released from the pair of rollers of the image forming apparatus **120** sandwiching the sheet P. Note that the sub-scan direction is the direction corresponding to the conveyance direction in which the sheet is conveyed.

[0035] Accordingly, with a configuration in which a sheet is conveyed from the image forming apparatus **120** with a different housing from the image inspection apparatus **110**, elongation and shortening of the image for reading may occur. When print quality is inspected on the basis of a comparison between such an image for reading and a correct image (for example, RIP image), the inspection accuracy may be decreased. However, because the present embodiment has the following configuration, a decrease to inspection accuracy can be suppressed.

[0036] By inverting the sign of the profile **501** illustrated in FIG. **3**, a correction profile may be created, and thus the profile **501** may be used as a correction profile.

Correction Profile Creation

[0037] FIG. **4** is a diagram illustrating the sheet P (may also be referred to as a test chart or a predetermined image) on which a test image **400** is formed. The test image **400** includes a predetermined reference pattern. The image inspection apparatus **110** reads the image printed on the sheet P and creates an image for reading (image for inspection). The predetermined reference pattern allows the image inspection apparatus **110** to detect distortion in the image for reading (image for inspection). An example of such a reference pattern includes a pattern (parallel line

pattern) including a plurality of lateral lines arranged at predetermined intervals.

[0038] For example, when the user inputs a predetermined instruction via the console unit **260**, the CPU **201** transmits the image data of the test image **400** and a print command to the internal controller **210b**. The predetermined instruction is an instruction (instruction for forming a test chart) for creating a correction profile. The image data of the test image **400** may be stored in the ROM area of the memory **202**. The CPU **211b** generates an image signal corresponding to the image data and outputs the image signal to the exposure device **23**. Furthermore, the CPU **211b** controls the image forming apparatus **120** and causes the image forming apparatus **120** to form the test image **400** on the sheet P.

[0039] Also, when the user inputs a predetermined instruction via the console unit **260**, the CPU **201** controls the internal controller **210a** and causes the internal controller **210a** to read the sheet P on which the test image **400** is formed. The CPU **211a** of the internal controller **210a** reads the sheet P on which the test image **400** is formed in accordance with the read command transmitted from the CPU **201** and transmits the reading result (image data) to the CPU **201**.

[0040] The CPU **201** creates the correction profile based on the reading result of the test image **400**. For example, the CPU **201** calculates the misalignment amount of the position in the sub-scan direction of the image (image for inspection) representing the reading result of the test image **400** against the test image **400** on the basis of the difference between the reading result of the test image **400** and the original data of the test image **400**. The misalignment amount corresponds to the value on the vertical axis illustrated in FIG. 3, for example.

[0041] The CPU **201** creates the correction profile to use to reduce the calculated misalignment amount. For example, the CPU **201** associates a value corresponding to the value on the vertical axis illustrated in FIG. 3 with an inverted sign with a position in the sub-scan direction and stores these in the memory **202**.

[0042] In the case of inspecting a printed image, the CPU **201** uses the correction profile described above and corrects the image data corresponding to the image for inspection and performs inspection on the basis of the post-correction image data and the image data of the correct image.

[0043] FIG. 5 is a flowchart illustrating a method of the external controller **200** of the present embodiment creating a correction profile. The processing of the flowchart illustrated in FIG. 5 is executed by the CPU **201** of the external controller **200**.

[0044] In step **S101**, the CPU **201** determines whether or not an instruction to create a correction profile has been input via the console unit **260**. When an instruction has been input, the CPU **201** advances the processing to step **S102**.

[0045] In step **S102**, the CPU **201** prints a test image. For example, the CPU **201** transmits the image data of the test image **400** and a print command therefor to the internal controller **210b**. The CPU **211b** generates an image signal corresponding to the image data and outputs the image signal to the exposure device **23**. Furthermore, the CPU **211b** controls the image forming apparatus **120** and causes the image forming apparatus **120** to form the test image **400** on the sheet P. Note that the instruction to create a correction profile may be input via an external apparatus (PC, smart phone, tablet, or the like) of the image forming system **100**.

[0046] In step **S103**, the CPU **201** acquires the reading result of the test image. For example, the CPU **201** controls the internal controller **210a** and causes the internal controller **210a** to read the sheet P on which the test image **400** is formed. The CPU **211a** of the internal controller **210a** reads the sheet P on which the test image **400** is formed in accordance with the read command transmitted from the CPU **201** and transmits the reading result (image data) to the CPU **201**.

[0047] In step **S104**, the CPU **201** creates the correction profile based on the reading result of the test image **400** using the method described above.

[0048] The CPU **201** performs image inspection on the basis of the image data corrected using the correction profile and the image data of the correct image.

[0049] As described above, in the present embodiment, when an instruction to create a correction

profile is input, the test image **400** is formed on the sheet P, and the test image **400** is then read by the image inspection apparatus **110**. The CPU **201** creates the correction profile based on the reading result of the test image **400** using the method described above. In the case of image inspection, the external controller **200** corrects the image data acquired from the image inspection apparatus **110** using the correction profile. As a result, elongation and shortening in the image for reading is reduced. The elongation and shortening is caused by the conveyance speed of the sheet P when the sheet P is discharged by the image forming apparatus **120** being different from the conveyance speed of the sheet P when the sheet P is taken in and conveyed by the image inspection apparatus **110**. Inspection of the image is performed by comparing the post-correction image data and the image data of the correct image. As a result, a decrease in the print quality inspection accuracy can be suppressed.

[0050] The image inspection may be performed by the internal controller **210a**. In this case, the CPU **211a** receives the image data of the correct image from the external controller **200** via the image communication line **222a**. The CPU **211a** corrects the image data of the image for inspection using the correction profile described below. Also, the CPU **211a** performs image inspection on the basis of the post-correction image data and the image data of the received correct image. In this manner, the image inspection may be performed by the external controller **200** or performed by the internal controller **210a**.

[0051] In the present embodiment, the image data corresponding to the image for inspection is corrected via the correction profile and inspection is performed on the basis of the post-correction image data and the image data corresponding to the correct image. However, this is merely one example. For example, the image data corresponding to the correct image may be corrected via the correction profile and inspection may be performed on the basis of the post-correction image data and the image data corresponding to the image for inspection.

[0052] A configuration in which the configuration of the image inspection apparatus **110** is included in the image forming apparatus **120** (in other words, the image reading units **11a** and **11b** are provided in the image forming apparatus **120**) may be applied to the present embodiment. In other words, the motor M1 that drives the conveyance roller **4a** functioning as a conveyance unit included in the discharge conveying apparatus **4** and the motor M2 that drives the inlet roller **15** may be different. In a case in which a plurality of motors are used, elongation and shortening in the image as described above may occur. Thus, by applying the present embodiment, the image data is appropriately corrected and the accuracy of image inspection is improved.

Second Embodiment

[0053] The configuration of the image forming system **100** of the second embodiment is similar to the configuration described for the first embodiment. Thus, common components will be omitted from the description.

[0054] In the image forming system **100**, for example, an inserter apparatus may be provided between the image inspection apparatus **110** and the image forming apparatus **120**. As illustrated in FIG. **6**, in the image forming system **100**, for example, a large capacity loading apparatus (hereinafter, referred to as a stacker **140**) is provided between the image inspection apparatus **110** and the post-processing apparatus **130**. The stacker **140** includes a housing **54**, conveying units **40a** and **40b**, an upper tray **45**, and a large capacity tray **43**. The conveying unit **40a** includes a plurality of conveyance rollers **41** and path switching mechanisms **42a** and **42b**. The path switching mechanism **42a** can guide the sheet P to the large capacity tray **43**. The path switching mechanism **42b** guides the sheet P not guided to the large capacity tray **43** to the upper tray **45** or the conveying unit **40b**. The conveying unit **40b** guides the sheet P to the post-processing apparatus **130**.

[0055] For example, the apparatus connected to the upstream side of the image inspection apparatus **110** when an instruction for creating a correction profile is input may be the same as the apparatus connected to the upstream side of the image inspection apparatus **110** when a correction profile was last created. In this case, the degree of elongation and shortening in the image as

illustrated in FIG. 3 will not significantly change or the change amount will be small. In such a state, the correction profile last created can be used. Irrespective of this, when an instruction to create a correction profile is input, the test image **400** is formed on the sheet P. In other words, the sheet P may be needlessly consumed (waste paper may be produced). However, when the apparatus on the upstream side or the apparatus on the downstream side of the image inspection apparatus **110** is changed, the tendency in variation illustrated in FIG. 3 changes. Here, with the present embodiment with the configuration described below, the production of waste paper can be suppressed, and a decrease in the print quality inspection accuracy can be suppressed.

[0056] FIG. 7 is a flowchart illustrating a method of the external controller **200** of the present embodiment creating a correction profile. The processing of the flowchart illustrated in FIG. 7 is executed by the CPU **201** of the external controller **200**.

[0057] In step S201, the CPU **201** determines whether or not an instruction to create a correction profile has been input via the console unit **260**. When an instruction has been input, the CPU **201** advances the processing to step S202.

[0058] In step S202, the CPU **201** detects the combination of apparatuses connected to the image inspection apparatus **110** and determines whether or not the combination has been changed. For example, the CPU **201** may, in order, request identification information or connection position information from the image inspection apparatus **110**, the image forming apparatus **120**, and the post-processing apparatus **130** via the serial communication lines **221a**, **221b**, and **221c**.

Alternatively, by the CPU **201** transmitting a query signal and acquiring identification information in order from the apparatus located upstream in the conveyance direction of the sheet P, which apparatus is connected at which position may be identified. Also, each apparatus may be provided with a sensor that recognizes the apparatus connected to the upstream side and a sensor that recognizes the apparatus connected to the downstream side. The CPU **201** causes each apparatus to report their identification information, the identification information of the apparatus on the upstream side, and the identification information of the apparatus on the downstream side. In this case, the image forming apparatus **120** reports the identification information of the image forming apparatus **120** and the identification information of the image inspection apparatus **110**. The image inspection apparatus **110** reports the identification information of the image forming apparatus **120**, the identification information of the image inspection apparatus **110**, and the identification information of the post-processing apparatus **130**. The post-processing apparatus **130** reports the identification information of the post-processing apparatus **130** and the identification information of the image inspection apparatus **110**. An apparatus that reports two items of identification information is an apparatus located furthest upstream or an apparatus located furthest downstream in the image forming system **100**. An apparatus that reports three items of identification information is an apparatus connected to an apparatus on the upstream side and an apparatus on the downstream side. The CPU **201** identifies the connection position of the apparatuses according to this rule for deciding.

[0059] For example, management information indicating the combination of apparatuses detected when the last correction profile was created may be held in the ROM area of the memory **202**. By comparing the currently detected combination and the combination stored in the memory **202**, the CPU **201** may determine whether or not the combination of apparatuses connected to the image inspection apparatus **110** has changed. In a case in which the combination has changed, the CPU **201** advances the processing to step S203. In a case in which the combination has not changed, the CPU **201** ends the processing of the flowchart. Note that in a case in which the last combination is not stored in the memory **202** or a profile is not stored in the memory **202**, the CPU **201** advances the processing to step S203. This is because such cases are analogous to a case in which the combination has changed.

[0060] In step S203, the CPU **201** prints the test image **400**. For example, the CPU **201** transmits the image data of the test image **400** and a print command therefor to the internal controller **210b**.

The CPU **211b** generates an image signal corresponding to the image data and outputs the image signal to the exposure device **23**. Furthermore, the CPU **211b** controls the image forming apparatus **120** and causes the image forming apparatus **120** to form the test image **400** on the sheet P. Note that the instruction to create a correction profile may be input via an external apparatus (PC, smart phone, tablet, or the like) of the image forming system **100**.

[0061] In step **S204**, the CPU **201** controls the internal controller **210a** and causes the internal controller **210a** to read the sheet P on which the test image **400** is formed. The CPU **211a** of the internal controller **210a** reads the sheet P on which the test image **400** is formed in accordance with the read command transmitted from the CPU **201** and transmits the reading result (image data) to the CPU **201**.

[0062] In step **S205**, the CPU **201** updates the correction profile or creates a new correction profile using the method described for the first embodiment on the basis of the reading result of the test image **400**.

[0063] The correction profile may be the profile **501** illustrated in FIG. **3** with the sign inverted. Thus, the profile **501** may be used as a correction profile.

[0064] In the present embodiment, the CPU **201** stores the management information for managing the relationship between the profile **501** and the combination of the apparatuses connected to the image inspection apparatus **110** in the memory **202**. An example of the management information is illustrated in FIG. **8A**. In this example, for an upstream apparatus i, “printer A” representing the identification information of the image forming apparatus **120** is stored. In the present embodiment, because the image inspection apparatus **110** is directly connected to the image forming apparatus **120**, nothing is present in an upstream apparatus ii. Also, in a similar manner, nothing is present in a downstream apparatus ii. For a downstream apparatus i, “post-processing apparatus” representing the identification information of the post-processing apparatus **130** is stored. The identification information does not need to be a name/title and may be a combination of numerical values, a combination of numerical values and letters of the alphabet, and the like.

[0065] Here, there are advantages in storing management information **600** in the memory **202**. For example, even in a case in which the current combination and the last combination are different, there are cases in which the current combination and the combination used previous to last match. In this case, the CPU **201** can skip updating the profile **501**. In other words, in a case in which the profile **501** associated with the current combination is already acquired, updating the profile **501** is skipped. Accordingly, the waiting time of the user is shortened.

[0066] As described above, the apparatus (current apparatus) connected to the image inspection apparatus **110** when an instruction for creating a correction profile is input may be the same as the apparatus (last apparatus) connected to the image inspection apparatus **110** when the last correction profile was created. In this case, creation of a correction profile is not performed. In other words, the last correction profile created is used in the image inspection. Because creation of the correction profile is skipped, the waiting time for the user is reduced.

[0067] However, the current apparatus may be different from the last apparatus. In this case, creation of a correction profile is performed.

[0068] As a result, producing waste paper can be suppressed, and a decrease in the print quality inspection accuracy can be suppressed.

[0069] In a case in which the current apparatus is the same as the last apparatus, a notification indicating that correction profile creation will not be performed may be displayed on the display panel provided on the console unit **260**. This allows the user to come to know that correction profile creation will not be performed. Even when there is an instruction to create a correction profile from the user, in a case in which the test chart is not printed, the user may mistakenly think that the image forming system **100** malfunctioned. In the present embodiment, because the user is notified that correction profile creation will not be performed, the user will not be mistaken.

Examples of Other Combinations

[0070] FIG. 8B is an example of the management information **600** with the profile **501** for the combination illustrated in FIG. 6 newly added. A profile number of PF2 is allocated to the profile **501** for the combination illustrated in FIG. 6. This allows for a distinction to be made with a profile number PF1 of the profile **501** for the combination illustrated in FIG. 1. Also, according to FIG. 8B, the identification information of the stacker **140** is listed as the downstream apparatus i, and the identification information of the post-processing apparatus **130** is listed as the downstream apparatus ii.

[0071] In this manner, when the combination of the apparatuses connected to the image inspection apparatus **110** is changed, the profile **501** is created or selected according to this combination. Thus, the quality of the printed image can be inspected with greater accuracy in accordance with the combination of the apparatuses connected to the image inspection apparatus **110**.

Technical Ideas Derived from Examples

Perspective A1

[0072] The image forming system **100** is an example of an image inspection system. The image forming apparatus **120** is an example of an image forming apparatus that forms a printed image corresponding to an original image on a sheet. The image inspection apparatus **110** is an example of an image inspection apparatus that reads the sheet and acquires an image for inspection corresponding to the printed image. The memory **202** is an example of a storage unit that stores in advance correction data for correcting distortion in the image for inspection for the printed image generated at the image inspection apparatus. The CPU **201** is an example of a correction unit that corrects the image for inspection on the basis of the correction data. The CPU **201** is an example of an inspection unit that inspects the print quality of the printed image on the basis of the original image and the image for inspection. The CPU **201** is an example of a detection unit that detects whether or not the sheet conveying apparatus connected to the upstream side or the downstream side of the image inspection apparatus in the sheet conveyance direction has changed. The CPU **201** is an example of an updating unit that updates the correction data when the CPU **201** detects that the sheet conveying apparatus connected to the upstream side or the downstream side of the image inspection apparatus has changed. In this manner, according to the present embodiment, when the sheet conveying apparatus connected to the upstream side or the downstream side of the image inspection apparatus is changed, the correction data is updated. Thus, according to the present embodiment, the waiting time for the user is decreased, and image inspection can be performed with accuracy. In FIG. 2, the CPU **201** is installed in the external controller **200**. However, this is not required. The CPU **201** may be installed in the image forming apparatus **120**, the image inspection apparatus **110**, or the post-processing apparatus **130**. In such cases, instead of the description above reading as CPU **201** and memory **202**, it would read as CPU **211** and memory **212**. Note that the external controller **200** may function as an image inspection apparatus including a storage unit, a correction unit, an inspection unit, a detection unit, and an updating unit.

Perspective A2

[0073] The memory **202** may function as a storage unit that stores in advance the identification information of the sheet conveying apparatuses included in the image inspection system. The CPU **201** may function as a collection unit that collects the identification information of the sheet conveying apparatuses included in the image inspection system when a predetermined collection start condition is satisfied. The CPU **201** compares the identification information of the sheet conveying apparatuses stored by the storage unit and the identification information of the sheet conveying apparatuses included in the image inspection system collected by the collection unit. Also, the CPU **201** may detect whether or not the sheet conveying apparatus connected to the upstream side or the downstream side of the image inspection apparatus in the sheet conveyance direction has changed on the basis of the comparison result. This allows for changes to the sheet conveying apparatuses to be reliably detected.

Perspective A3

[0074] The memory **202** may function as a storage unit that stores in advance the position information indicating the connection position of the sheet conveying apparatuses included in the image inspection system. The CPU **201** may function as a collection unit that collects the position information indicating the connection position of the sheet conveying apparatuses included in the image inspection system when a predetermined collection start condition is satisfied. The CPU **201** compares the position information of the sheet conveying apparatuses stored by the storage unit and the position information of the sheet conveying apparatuses included in the image inspection system collected by the collection unit. The CPU **201** may detect whether or not the sheet conveying apparatus connected to the upstream side or the downstream side of the image inspection apparatus in the sheet conveyance direction has changed.

Perspective A4

[0075] The memory **202** may function as a storage unit that stores in advance the identification information of the sheet conveying apparatus connected to the upstream side of the image inspection apparatus and the identification information of the sheet conveying apparatus connected to the downstream side of the image inspection apparatus. When a predetermined collection start condition is satisfied, the CPU **201** may function as a collection unit that collects the identification information of the sheet conveying apparatus connected to the upstream side of the image inspection apparatus and the identification information of the sheet conveying apparatus connected to the downstream side of the image inspection apparatus. The CPU **201** compares the two combinations. The first combination is a combination of the identification information of the sheet conveying apparatus connected to the upstream side of the image inspection apparatus and the identification information of the sheet conveying apparatus connected to the downstream side of the image inspection apparatus collected by the collection unit and stored by the storage unit. The second combination is a combination of the identification information of the sheet conveying apparatus connected to the upstream side of the image inspection apparatus and the identification information of the sheet conveying apparatus connected to the downstream side of the image inspection apparatus collected by the collection unit. The CPU **201** may detect whether or not the sheet conveying apparatus connected to the upstream side or the downstream side of the image inspection apparatus in the sheet conveyance direction has changed on the basis of the comparison result.

Perspective A5

[0076] The console unit **260** may function as an accepting unit that accepts input of information indicating whether or not the combination of the sheet conveying apparatuses included in the image inspection system has changed. The CPU **201** may detect whether or not the sheet conveying apparatus connected to the upstream side or the downstream side of the image inspection apparatus in the sheet conveyance direction has changed on the basis of the information accepted by the accepting unit.

Perspective A6

[0077] The predetermined collection start condition may be that the image inspection system is supplied with power from an external power supply and activates. The predetermined collection start condition may be an instruction to start the collection processing for information relating to the sheet conveying apparatuses. The instruction may be input via the console unit **260**, for example.

Perspective A7

[0078] The CPU **201** causes the image forming apparatus to form a printed image corresponding to the test image on a sheet and causes the image inspection apparatus to read the printed image formed on the sheet. Also, the CPU **201** may update the correction data on the basis of the reading result of the printed image corresponding to the test image and the test image. Accordingly, correction data is created that can reduce the distortion from reading the printed image. The test image may include a plurality of line drawings intersecting the sheet conveyance direction and

arranged at even intervals. As illustrated in FIG. 4, the plurality of line drawings may be orthogonal to the sheet conveyance direction.

[0079] Accordingly, the detection accuracy of elongation or shortening of the image for inspection in the sheet conveyance direction is improved.

Perspective A8

[0080] One of the sheet conveying apparatuses connected to the upstream side of the image inspection apparatus may be the image forming apparatus. Also, one or more sheet conveying apparatuses may be connected between the image inspection apparatus and the image forming apparatus.

Perspective A9

[0081] The distortion of the image for inspection includes at least one of position misalignment, elongation, or shortening in the image for inspection with respect to the printed image. Distortion in the image for inspection may be caused by a variation in the rotational speed of the rotary bodies involved in conveying a sheet. Such distortion is distortion that may occur when the printed image is read.

Perspective A10

[0082] The sheet conveying apparatus connected to the downstream side of the image inspection apparatus may include a sorting function. For example, the discharge tray **34b** may function as a first loading unit for loading sheets on which the printed image is formed that has been determined to satisfy the predetermined print quality by the inspection unit. The discharge tray **34a** may function as a second loading unit for loading sheets on which the printed image is formed that has been determined to not satisfy the predetermined print quality by the inspection unit.

Perspective A11

[0083] The CPU **201** may detect the combination of the sheet conveying apparatuses connected to the upstream side and the downstream side of the image inspection apparatus in the sheet conveyance direction. The CPU **201** may determine whether or not the correction data corresponding to the combination detected by the detection unit is stored in the storage unit. The correction data corresponding to the combination detected by the detection unit may not be stored in the storage unit. In this case, the CPU **201** may function as a creation unit that creates the correction data corresponding to the combination detected by the detection unit. The CPU **201** may associate the correction data with the combination detected by the detection unit and store the correction data in the storage unit. However, correction data corresponding to the combination detected by the detection unit may already be stored in the storage unit. In this case, the CPU **201** may skip creating the correction data.

Perspective B1

[0084] The image forming system **100** may include [0085] an image forming unit (for example, the image forming apparatus **120**) configured to form an image corresponding to first image data input into the image forming system on a sheet, [0086] a first conveyance unit (for example, the conveyance roller **4a**) configured to convey the sheet conveyed from the image forming unit, [0087] a first motor (for example, the motor **M1**) configured to drive the first conveyance unit, [0088] a second conveyance unit (for example, the inlet roller **15**) configured to convey the sheet conveyed from the first conveyance unit, [0089] a second motor (for example, the motor **M2**) configured to drive the second conveyance unit, [0090] a reading unit (for example, the image reading unit **11a**) configured to read an image on the sheet being conveyed by the second conveyance unit, and [0091] a processor (for example, the CPU **201**, **211a**, **211b**, and **211c**). The processor may include one or more processors/processor cores/processing circuitries.

[0092] The processor may be configured to [0093] generate second image data representing a result of the reading unit reading an image on the sheet; [0094] correct the second image data using correction data, the correction data being data obtained on the basis of image data obtained by the reading unit reading a predetermined image formed on a sheet by the image forming unit and image

data corresponding to the predetermined image; and [0095] inspect, on the basis of the corrected second image data and the first image data, an image formed on the sheet on the basis of the first image data.

[0096] In this manner, the image for inspection may be corrected.

Perspective B2

[0097] In the image forming system according to perspective B1, the image forming unit, the first conveyance unit, and the first motor may be provided in a first housing (for example, the housing **52**); and the second conveyance unit, the second motor, and the reading unit may be provided in a second housing (for example, the housing **51**) different from the first housing.

Perspective B3

[0098] In the image forming system according to perspective B1, when the processor receive an instruction to create the correction data, the processor may cause the image forming unit to form a predetermined image on a sheet and may cause the reading unit to read an image on a sheet on which the predetermined image is formed.

[0099] Also, the processor may create the correction data by comparing image data obtained by the reading unit reading an image on the sheet on which the predetermined image is formed and image data corresponding to the predetermined image.

Perspective B4

[0100] In the image forming system according to perspective B1, the image forming unit, the first conveyance unit, and the first motor may be provided in a first housing (for example, the housing **52**), and the second conveyance unit, the second motor, and the reading unit may be provided in a second housing (for example, the housing **51**) different from the first housing.

[0101] The image forming system **100** may further include a storage unit (for example, the memory **202** and **212a**) that stores the correction data and first information relating to a housing connected to the second housing when correction data is created. The processor may correct the second image data using the correction data stored in the storage unit.

[0102] When the processor receives an instruction to create the correction data, the processor may acquire second information relating to a housing connected to the second housing.

[0103] In a case in which the first information and the second information are different, the processor may execute creation processing to create the correction data. Here, the creation processing may include (i) the processor causing the image forming unit to form the predetermined image on the sheet and causing the reading unit to read an image on the sheet on which the predetermined image is formed, (ii) the processor creating the correction data by comparing image data obtained by the reading unit reading an image on the sheet on which the predetermined image is formed and image data corresponding to the predetermined image, and (iii) the processor storing the created correction data in the storage unit and storing the second information as the first information in the storage unit.

[0104] In a case in which the first information and the second information are identical, the processor may not execute the creation processing.

Perspective B5

[0105] The image forming system **100** according to perspective B4 further includes a notification unit (for example, the console unit **260**) configured to notify of information indicating that the creation processing has not been executed in a case in which the first information and the second information are identical.

Perspective B6

[0106] In the image forming system **100** according to perspective B1, the predetermined image may include a plurality of line drawings intersecting a conveyance direction of the sheet and arranged at even intervals in the conveyance direction of the sheet.

Perspective B7

[0107] The image forming system **100** may include [0108] an image forming unit configured to

form an image corresponding to first image data input into the image forming system on a sheet, [0109] a first conveyance unit configured to convey the sheet conveyed from the image forming unit, [0110] a first motor configured to drive the first conveyance unit, [0111] a second conveyance unit configured to convey the sheet conveyed from the first conveyance unit, [0112] a second motor configured to drive the second conveyance unit, [0113] a reading unit configured to read an image on the sheet being conveyed by the second conveyance unit, and [0114] a processor. [0115] The processor may be configured to [0116] generate second image data representing a result of the reading unit reading an image on the sheet; [0117] correct the first image data using correction data, the correction data being data obtained on the basis of image data obtained by the reading unit reading a predetermined image formed on a sheet by the image forming unit and image data corresponding to the predetermined image; and [0118] inspect, on the basis of the corrected first image data and the second image data, an image formed on the sheet on the basis of the first image data.

[0119] Accordingly, the original image (also referred to as an image for comparison, a correct image, or a reference image) compared to the image for inspection may be corrected.

Perspective B8

[0120] In the image forming system **100** according to perspective B7, the image forming unit, the first conveyance unit, and the first motor may be provided in a first housing. The second conveyance unit, the second motor, and the reading unit may be provided in a second housing different from the first housing.

Perspective B9

[0121] In the image forming system according to perspective B7, [0122] when the processor receive an instruction to create the correction data, the processor may cause the image forming unit to form a predetermined image on a sheet and may cause the reading unit to read an image on a sheet on which the predetermined image is formed.

[0123] Also, the processor may create the correction data by comparing image data obtained by the reading unit reading an image on the sheet on which the predetermined image is formed and image data corresponding to the predetermined image.

Perspective B10

[0124] In the image forming system **100** according to perspective B7, [0125] the image forming unit, the first conveyance unit, and the first motor may be provided in a first housing. The second conveyance unit, the second motor, and the reading unit may be provided in a second housing different from the first housing.

[0126] The image forming system **100** may further include [0127] a storage unit (for example, the memory **202** and **212a**) that stores the correction data and first information relating to a housing connected to the second housing when correction data is created. The processor may correct the first image data using the correction data stored in the storage unit.

[0128] When the processor receives an instruction to create the correction data, the processor acquires second information relating to a housing connected to the second housing.

[0129] In a case in which the first information and the second information are different, the processor executes creation processing to create the correction data. Here, the creation processing includes the processor causing the image forming unit to form the predetermined image on the sheet and causing the reading unit to read an image on the sheet on which the predetermined image is formed, the processor creating the correction data by comparing image data obtained by the reading unit reading an image on the sheet on which the predetermined image is formed and image data corresponding to the predetermined image, and the processor storing the created correction data in the storage unit and storing the second information as the first information in the storage unit.

[0130] In a case in which the first information and the second information are identical, the processor does not execute the creation processing.

Perspective B11

[0131] The image forming system **100** according to perspective B10 may further include [0132] a notification unit (for example, the console unit **260**) configured to notify of information indicating that the creation processing has not been executed in a case in which the first information and the second information are identical.

Perspective B12

[0133] In the image forming system **100** according to perspective B7, the predetermined image may include a plurality of line drawings intersecting a conveyance direction of the sheet and arranged at even intervals in the conveyance direction of the sheet.

Perspective B13

[0134] The image inspection apparatus **110** includes [0135] a conveyance unit configured to convey a sheet conveyed from an image forming unit configured to form an image corresponding to input first image data on the sheet, [0136] a reading unit configured to read an image on the sheet being conveyed by the conveyance unit, and [0137] a processor (for example, the CPU **211a**). [0138] The processor is configured to [0139] generate second image data representing a result of the reading unit reading an image on the sheet; [0140] correct the second image data using correction data, the correction data being data obtained on the basis of image data obtained by the reading unit reading a predetermined image formed on a sheet by the image forming unit and image data corresponding to the predetermined image; and [0141] inspect, on the basis of the corrected second image data and the first image data, an image formed on the sheet on the basis of the first image data.

Perspective B14

[0142] The image inspection apparatus **110** includes [0143] a conveyance unit configured to convey a sheet conveyed from an image forming unit configured to form an image corresponding to input first image data on the sheet, [0144] a reading unit configured to read an image on the sheet being conveyed by the conveyance unit, and [0145] a processor.

[0146] The processor is configured to [0147] generate second image data representing a result of the reading unit reading an image on the sheet; [0148] correct the first image data using correction data, the correction data being data obtained on the basis of image data obtained by the reading unit reading a predetermined image formed on a sheet by the image forming unit and image data corresponding to the predetermined image; and [0149] inspect, on the basis of the corrected first image data and the second image data, an image formed on the sheet on the basis of the first image data.

OTHER EMBODIMENTS

[0150] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an

optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0151] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0152] This application claims the benefit of Japanese Patent Application No. 2021-166916, filed Oct. 11, 2021 and Japanese Patent Application No. 2022-129003, filed Aug. 12, 2022, which are hereby incorporated by reference herein in their entirety.

Claims

1-14. (canceled)

15. An image forming system comprising: an image forming unit configured to form an image on a sheet; a reading unit configured to read the image formed on the sheet; and one or more processors configured to generate read image data corresponding to a read image obtained by reading the image on the sheet by the reading unit, wherein in a case where an instruction to generate correction data for correcting the read image data is received, (1) the image forming unit forms a test image on the sheet, (2) the reading unit reads the test image formed on the sheet, and (3) the one or more processors generate the correction data for correcting the read image data based on the read image of the test image, wherein in a case where inspection of the image formed on the sheet is performed, (1) the image forming unit forms a first image on the sheet corresponding to input first image data, (2) the reading unit reads the first image formed on the sheet, and (3) the one or more processors generate first read image data corresponding to the first image, and correct the first read image data using the correction data, and wherein the one or more processors inspect the first image formed on the sheet based on (1) the corrected first read image data and (2) the input first image data.

16. The image forming system according to claim 15, wherein the one or more processors generate the correction data by comparing (1) second read image data corresponding to the read image of the test image and (2) second image data corresponding to the test image.

17. The image forming system according to claim 15, wherein the test image includes a plurality of line images that are arranged at equal intervals in a conveyance direction in which the sheet is conveyed, and that intersect the conveyance direction.

18. An image forming system comprising: an image forming unit configured to form an image on a sheet; a reading unit configured to read the image formed on the sheet; and one or more processors configured to generate read image data corresponding to a read image obtained by reading the image on the sheet by the reading unit, wherein in a case where an instruction to generate correction data for correcting the read image data is received, (1) the image forming unit forms a test image on the sheet, (2) the reading unit reads the test image formed on the sheet, and (3) the one or more processors generate the correction data for correcting the read image data based on the read image of the test image, wherein in a case where inspection of the image formed on the sheet is performed, (1) the one or more processors correct input first image data using the correction data, (2) the image forming unit forms a first image corresponding to the corrected first image data on a sheet, (3) the reading unit reads the first image formed on the sheet, and (4) the one or more processors generate first read image data corresponding to the first image data, and wherein the one or more processors inspect the first image formed on the sheet based on (1) the corrected first image data and (2) the first read image data.

19. The image forming system according to claim 18, wherein the one or more processors generate the correction data by comparing (1) second read image data corresponding to the read image of the test image and (2) second image data corresponding to the test image.

20. The image forming system according to claim 18, wherein the test image includes a plurality of line images that are arranged at equal intervals in a conveyance direction in which the sheet is conveyed, and that intersect the conveyance direction.
